MLOps Assignment 2

Tracking a Pre-trained Sentiment Model with MLflow

1. Aim

To track a pre-trained sentiment analysis model using MLflow by logging parameters, metrics, and artifacts, comparing multiple experiment runs, and identifying the best-performing run.

2. Software Requirements

1. Python
2. MLflow
3. Hugging Face Transformers
4. PyTorch
5. Pandas
6. Visual Studio Code or any Python editor
7. Web browser

3. Theory

MLflow is an open-source platform used to manage and track machine learning experiments. It records parameters, metrics, and artifacts for each experiment run.

In this experiment, a pre-trained sentiment analysis model, distilbert-base-uncased-finetuned-sst-2-english, is used to classify text as positive or negative. The same model is executed with different batch sizes to compare accuracy, confidence, and inference latency.

The following information is tracked using MLflow:

- **Parameters:** Model name, batch size, and number of samples.
- **Metrics:** Accuracy, average confidence, and total inference latency.
- **Artifacts:** A CSV file containing actual labels, predicted labels, input text, and confidence scores.

4. Procedure

1. Install Python and the required libraries.
2. Create and activate a Python virtual environment.
3. Download the pre-trained sentiment analysis model.
4. Create a Python script named mlflow_lab.py.
5. Execute the model with batch sizes 1 and 8.
6. Log the parameters, metrics, and prediction results using MLflow.
7. Launch the MLflow user interface and compare the experiment runs.
8. Add a third experiment using batch size 4.
9. Observe the results and identify the run with the lowest inference latency.

6. Execution Commands

Step 1: Install the required packages

pip install mlflow transformers torch pandas

Step 2: Execute the program

python mlflow_lab.py

Step 3: Start the MLflow user interface

Open another terminal in the same project directory and execute:

mlflow ui

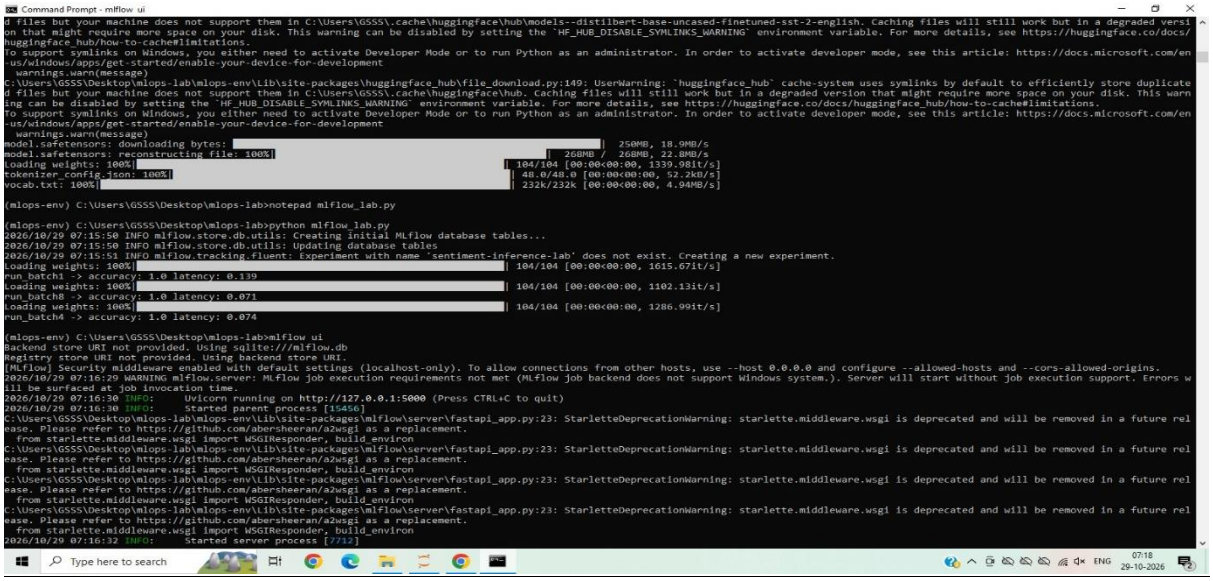
Open the following address in your browser:

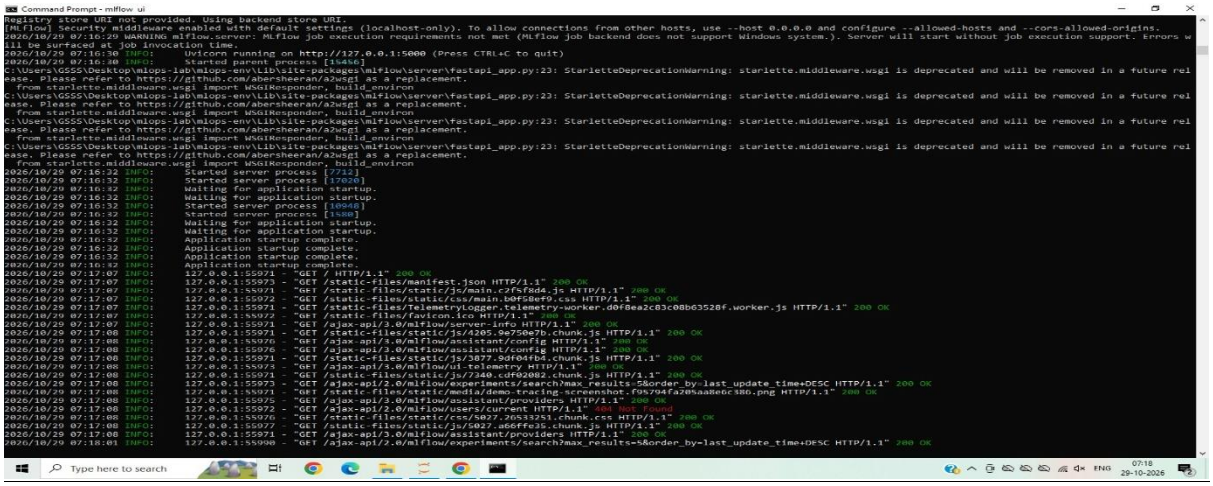
http://127.0.0.1:5000

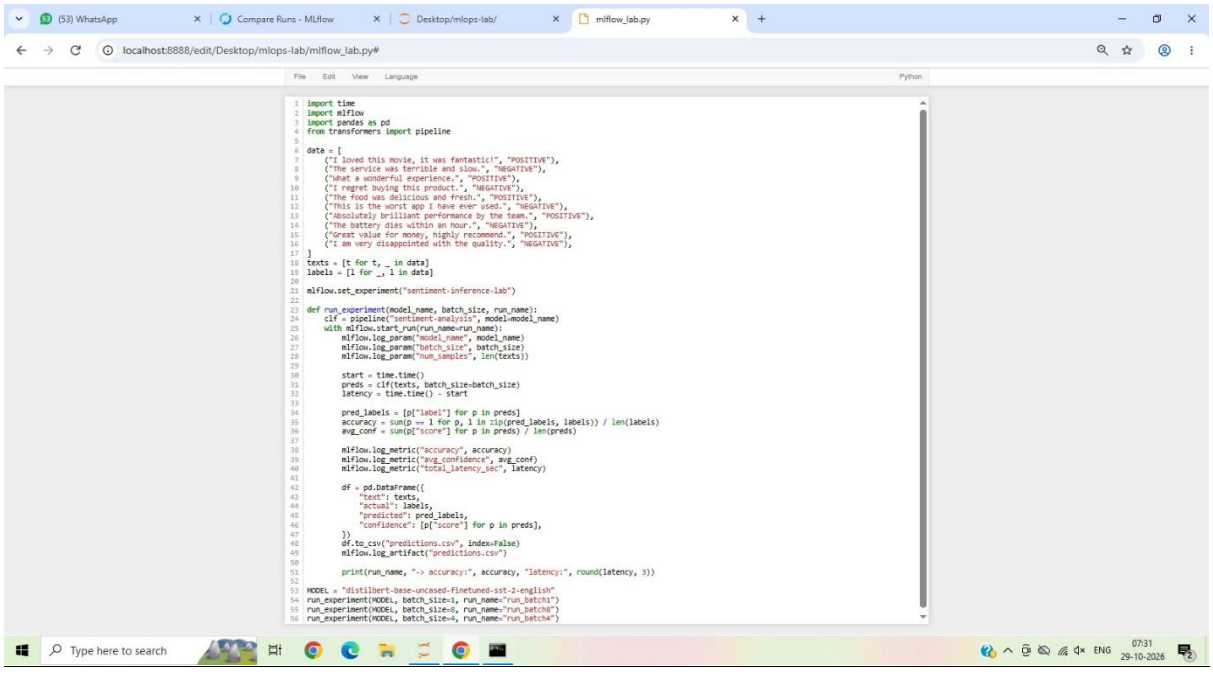
Step 4: Compare the experiment runs

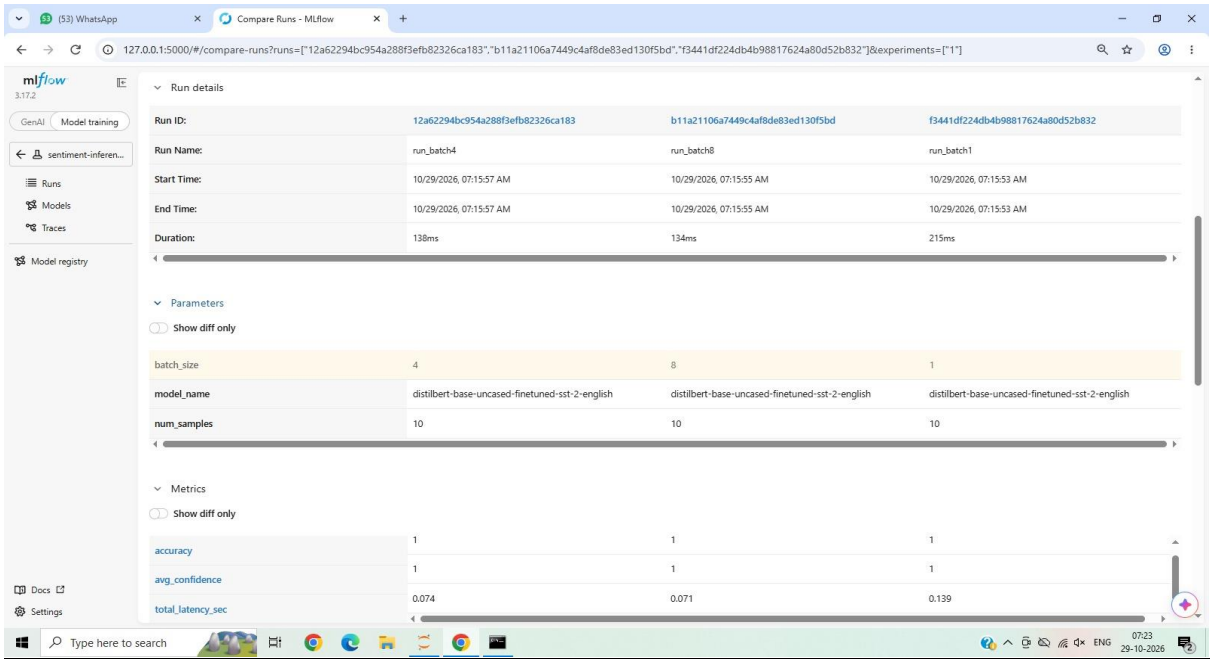
1. Open sentiment-inference-lab.
2. Select the three experiment runs.
3. Click **Compare**.
4. Open a run and view predictions.csv under Artifacts.

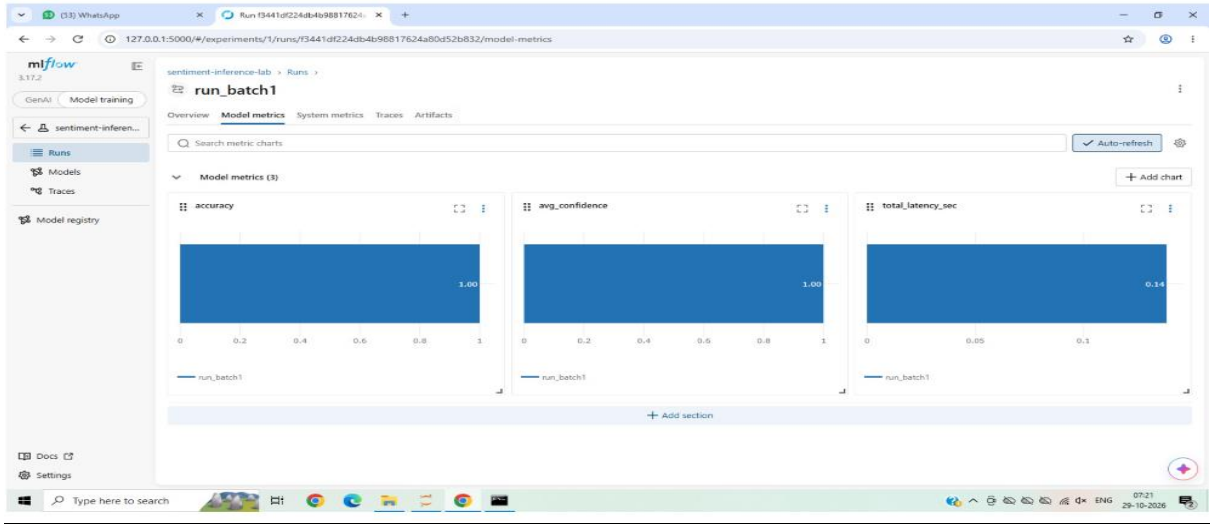
Outputs

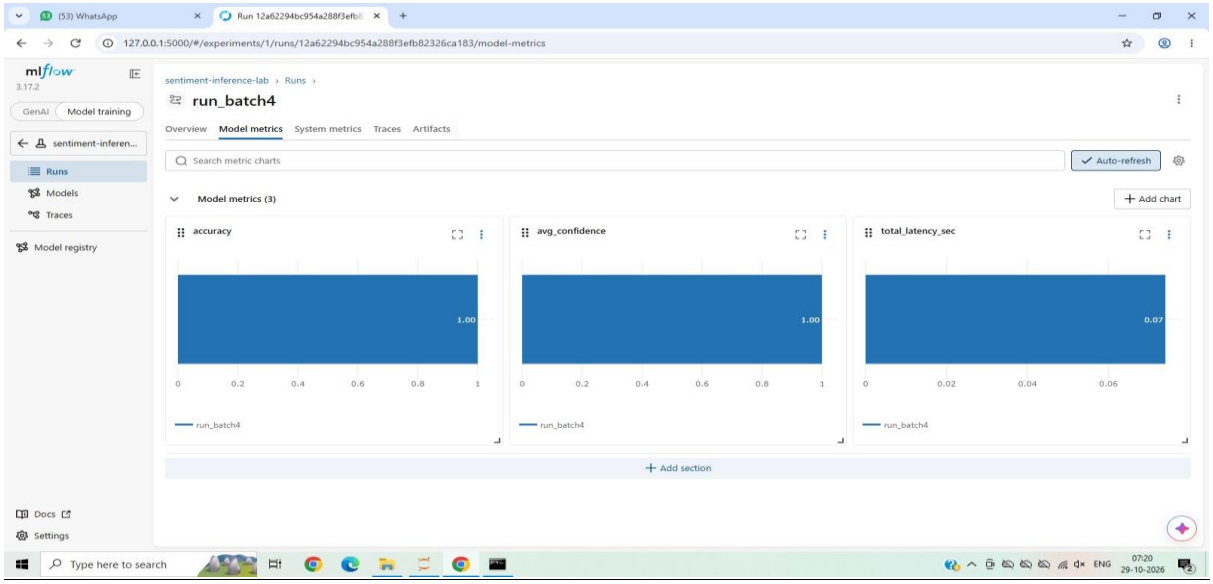

Command Prompt - mlflow ui
(mlops-env) C:\Users\GSSS\Desktop\mlops-lab>notepad mlflow_lab.py
(mlops-env) C:\Users\GSSS\Desktop\mlops-lab>python mlflow_lab.py
(mlops-env) C:\Users\GSSS\Desktop\mlops-lab>mlflow ui
Backend store URI not provided. Using sqlite:///mlflow.db
Registry store URI not provided. Using backend store URI.
[MLflow] Security middleware enabled with default settings (localhost-only). To allow connections from other hosts, use --host 0.0.0.0 and configure --allowed-hosts and --cors-allowed-origins.
Uvicorn running on http://127.0.0.1:5000 (Press CTRL+C to quit)
Started parent process [15456]
Started server process [7712]
Type here to search
ENG
07:18
29-10-2026

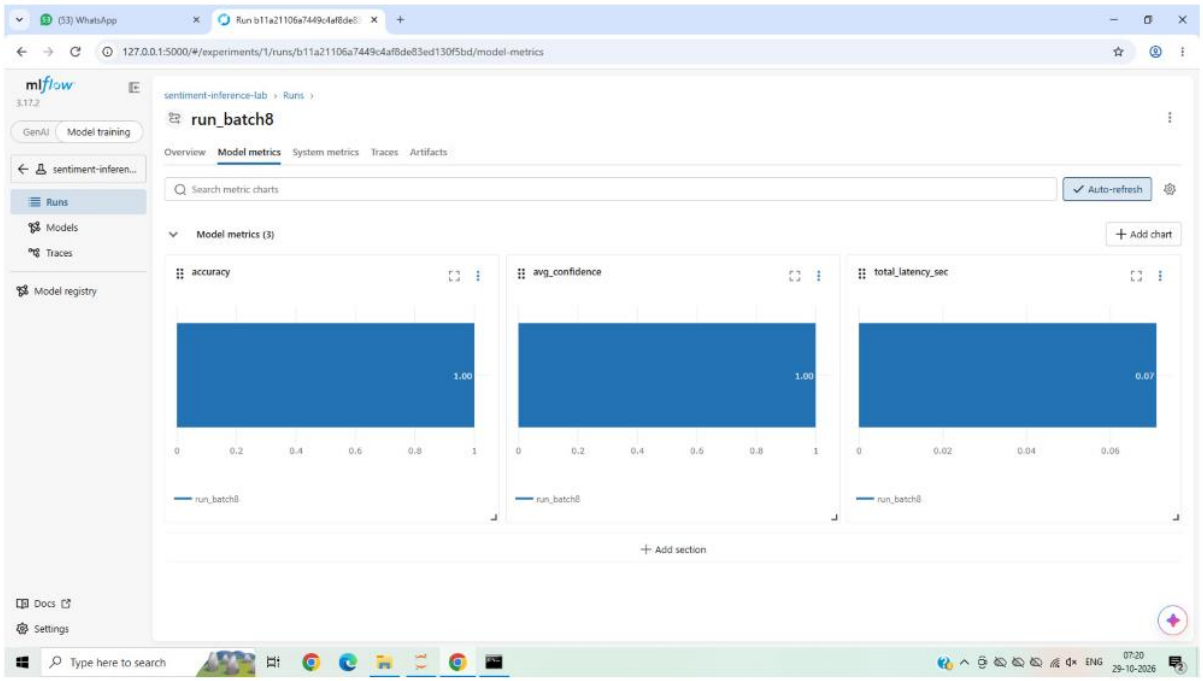

Command Prompt - mlflow ui
Registry store URI not provided. Using backend store URI.
[MLflow] Security middleware enabled with default settings (localhost-only). To allow connections from other hosts, use --host 0.0.0.0 and configure --allowed-hosts and --cors-allowed-origins.
Uvicorn running on http://127.0.0.1:5000 (Press CTRL+C to quit)
Started parent process [15456]
Started server process [7712]
Started server process [17020]
Waiting for application startup.
Started server process [10948]
Started server process [15480]
Application startup complete.
Type here to search
ENG
07:18
29-10-2026

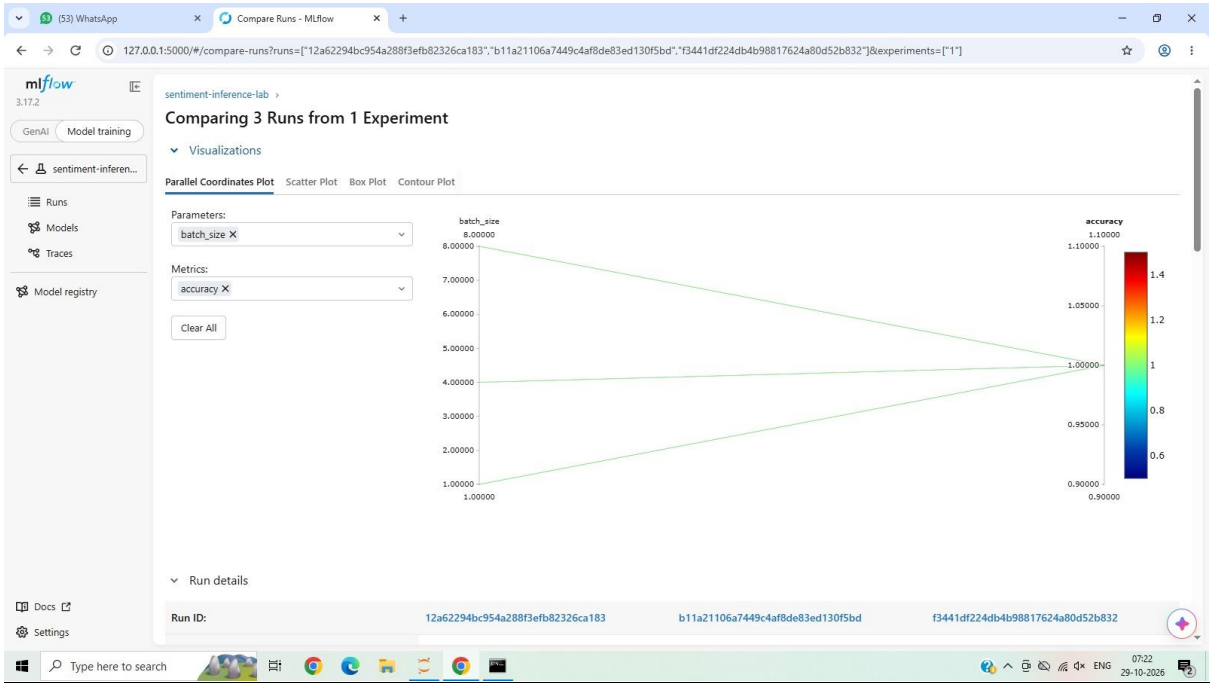

(53) WhatsApp
Compare Runs - MLflow
Desktop/mlops-lab/
mlflow_lab.py
localhost:8888/edit/Desktop/mlops-lab/mlflow_lab.py#
File Edit View Language
Python
import time
import mlflow
import pandas as pd
from transformers import pipeline
data = [
("I loved this movie, it was fantastic!", "POSITIVE"),
("The service was terrible and slow.", "NEGATIVE"),
("What a wonderful experience.", "POSITIVE"),
("I regret buying this product.", "NEGATIVE"),
("The food was delicious and fresh.", "POSITIVE"),
("This is the worst app I have ever used.", "NEGATIVE"),
("Absolutely brilliant performance by the team.", "POSITIVE"),
("The battery dies within an hour.", "NEGATIVE"),
("Great value for money, highly recommend.", "POSITIVE"),
("I am very disappointed with the quality.", "NEGATIVE"),
]
texts = [t for t, _ in data]
labels = [1 for _, l in data]
mlflow.set_experiment("sentiment-inference-lab")
def run_experiment(model_name, batch_size, run_name):
clf = pipeline("sentiment-analysis", model=model_name)
with mlflow.start_run(run_name=run_name):
mlflow.log_param("model_name", model_name)
mlflow.log_param("batch_size", batch_size)
mlflow.log_param("num_samples", len(texts))
start = time.time()
preds = clf(texts, batch_size=batch_size)
latency = time.time() - start
pred_labels = [p["label"] for p in preds]
accuracy = sum(p == l for p, l in zip(pred_labels, labels)) / len(labels)
avg_conf = sum(p["score"] for p in preds) / len(preds)
mlflow.log_metric("accuracy", accuracy)
mlflow.log_metric("avg_confidence", avg_conf)
mlflow.log_metric("total_latency_sec", latency)
df = pd.DataFrame({
"text": texts,
"actual": labels,
"predicted": pred_labels,
"confidence": [p["score"] for p in preds],
})
df.to_csv("predictions.csv", index=False)
mlflow.log_artifact("predictions.csv")
print(run_name, "-> accuracy:", accuracy, "latency:", round(latency, 3))
MODEL = "distilbert-base-uncased-finetuned-sst-2-english"
run_experiment(MODEL, batch_size=1, run_name="run_batch1")
run_experiment(MODEL, batch_size=8, run_name="run_batch8")
run_experiment(MODEL, batch_size=4, run_name="run_batch4")
Type here to search
ENG
07:31
29-10-2026


Compare Runs - MLflow
127.0.0.1:5000/#/compare-runs?runs=["12a62294bc954a288f3efb82326ca183","b11a21106a7449c4af8de83ed130f5bd","f3441df224db4b98817624a80d52b832"]&experiments=["1"]
Run details
Run ID: 12a62294bc954a288f3efb82326ca183 b11a21106a7449c4af8de83ed130f5bd f3441df224db4b98817624a80d52b832
Run Name: run_batch4 run_batch8 run_batch1
Start Time: 10/29/2026, 07:15:57 AM 10/29/2026, 07:15:55 AM 10/29/2026, 07:15:53 AM
End Time: 10/29/2026, 07:15:57 AM 10/29/2026, 07:15:55 AM 10/29/2026, 07:15:53 AM
Duration: 138ms 134ms 215ms
Parameters
Show diff only
batch_size 4 8 1
model_name distilbert-base-uncased-finetuned-sst-2-english distilbert-base-uncased-finetuned-sst-2-english distilbert-base-uncased-finetuned-sst-2-english
num_samples 10 10 10
Metrics
Show diff only
accuracy 1 1 1
avg_confidence 1 1 1
total_latency_sec 0.074 0.071 0.139


sentiment-inference-lab › Runs ›
run_batch1
Overview Model metrics System metrics Traces Artifacts
Search metric charts
Auto-refresh
Model metrics (3)
Add chart
accuracy
avg_confidence
total_latency_sec
1.00
1.00
0.14
run_batch1
Add section


sentiment-inference-lab › Runs ›
run_batch4
Overview Model metrics System metrics Traces Artifacts
Search metric charts
Auto-refresh
Model metrics (3)
Add chart
accuracy
avg_confidence
total_latency_sec
1.00
1.00
0.07
run_batch4
Add section


mlflow
3.17.2
GenAI
Model training
sentiment-inferen...
Runs
Models
Traces
Model registry
sentiment-inference-lab › Runs ›
run_batch8
Overview
Model metrics
System metrics
Traces
Artifacts
Search metric charts
Auto-refresh
Model metrics (3)
Add chart
accuracy
avg_confidence
total_latency_sec
1.00
1.00
0.07
run_batch8
Add section
Docs
Settings
Type here to search
ENG
07:20
29-10-2026


(53) WhatsApp
Compare Runs - MLflow
127.0.0.1:5000/#/compare-runs?runs=["12a62294bc954a288f3efb82326ca183","b11a21106a7449c4af8de83ed130f5bd","f3441df224db4b98817624a80d52b832"]&experiments=["1"]
mlflow
3.17.2
GenAI
Model training
sentiment-inferen...
Runs
Models
Traces
Model registry
sentiment-inference-lab ›
Comparing 3 Runs from 1 Experiment
Visualizations
Parallel Coordinates Plot
Scatter Plot
Box Plot
Contour Plot
Parameters:
batch_size
Metrics:
accuracy
Clear All
batch_size
accuracy
Run details
Run ID:
12a62294bc954a288f3efb82326ca183
b11a21106a7449c4af8de83ed130f5bd
f3441df224db4b98817624a80d52b832
Docs
Settings
Type here to search
ENG
07:22
29-10-2026